

MONT-ORFORD, QC, CanadaWMO: **716181**Lat: **45.312N**Lon: **72.242W**Elev: **2776**StdP: **13.28**Time Zone: **-5.00 (NAE)**Period: **94-07**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.0	-13.2	-25.0	1.5	-18.4	-18.7	2.2	-12.8	41.0	27.5	36.8	18.8	17.1	300	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	77.2	65.3	74.3	63.9	71.6	62.8	69.0	73.6	66.7	70.7	64.9	68.9	9.8	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.4	111.8	71.2	65.4	104.2	69.0	63.5	97.2	67.0	34.9	73.9	33.0	70.8	31.4	68.8	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 35.1	30.3	27.2	-22.1	82.8	5.8	3.4	-26.3	85.2	-29.7	87.2	-33.0	89.1	-37.2	91.5		
(5)			-23.2	72.1	5.5	3.1	-27.1	74.3	-30.3	76.1	-33.4	77.9	-37.4	80.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.5	10.4	12.9	21.8	34.4	47.8	58.8	61.7	60.5	53.1	40.2	29.2	17.8
	DBStd	20.59	13.86	12.27	11.56	9.41	8.69	7.89	5.77	6.19	7.55	9.22	9.50	10.57
	HDD50	5788	1228	1038	874	476	146	17	2	52	330	625	999	
	HDD65	10130	1693	1458	1339	918	536	215	133	165	366	769	1074	1464
	CDD50	1232	0	0	1	8	78	281	366	329	143	26	0	0
	CDD65	98	0	0	0	0	2	29	33	27	7	0	0	0
	CDH74	285	0	0	0	0	6	115	77	61	26	0	0	0
	CDH80	27	0	0	0	0	0	15	5	4	3	0	0	0
	WSAvg	13.6	14.6	15.1	14.1	14.2	13.6	11.9	11.8	11.4	13.8	14.6	13.9	13.8
	PrecAvg	46.5	3.5	2.6	3.1	3.4	3.8	4.5	5.2	4.6	4.0	4.3	3.7	3.9
	PrecMax	55.7	6.1	3.7	4.5	7.4	6.2	6.7	8.9	7.4	8.5	9.2	5.6	8.0
(15) Precipitation	PrecMin	35.4	1.7	1.4	1.5	0.4	1.9	2.8	2.2	1.3	1.9	1.7	1.8	2.0
	PrecStd	5.1	1.1	0.7	0.7	1.7	1.1	1.2	1.6	1.5	1.7	2.0	1.0	1.5
	DB	48.5	42.1	54.4	65.6	74.5	82.2	80.0	79.4	77.8	68.1	55.7	45.9	
	MCWB	48.3	37.9	49.9	54.2	58.7	70.5	70.3	67.3	64.1	58.5	53.5	43.2	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	40.1	37.3	46.1	58.7	69.3	77.3	75.9	75.3	70.6	62.2	50.2	41.0	
	2% MCWB	39.1	35.0	40.0	48.6	55.8	64.4	66.8	64.4	61.7	54.8	47.5	38.7	
	5% DB	32.2	33.0	41.6	53.1	64.7	73.6	73.1	72.1	66.6	58.0	46.9	36.3	
	5% MCWB	31.9	31.1	37.4	43.6	52.8	63.0	65.0	63.4	59.4	51.7	44.3	35.0	
	10% DB	27.8	29.5	37.4	47.9	60.6	70.3	70.4	69.4	63.1	53.5	43.4	32.1	
	10% MCWB	27.4	27.7	33.9	41.4	51.8	61.8	63.9	61.9	58.4	48.6	41.1	31.1	
	0.4% WB	46.9	40.3	50.2	57.8	63.4	72.3	72.5	69.8	66.2	61.3	53.7	45.3	
	0.4% MCDB	47.0	40.9	55.3	61.8	69.4	78.0	76.9	76.0	72.2	65.4	54.8	46.9	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	38.7	35.3	42.9	50.3	59.5	68.3	69.3	67.1	63.8	55.6	48.5	39.3	
	2% MCDB	39.1	36.7	44.9	56.0	64.7	72.8	73.2	71.7	68.3	60.6	49.2	39.9	
	5% WB	31.7	31.2	38.3	45.8	56.0	65.6	67.1	65.3	61.4	52.7	45.3	34.7	
	5% MCDB	31.9	32.8	40.3	50.8	61.5	69.8	70.8	69.5	65.4	57.1	46.4	36.1	
	10% WB	27.1	27.9	33.9	41.9	53.3	63.1	65.1	63.6	59.2	49.5	41.0	30.7	
	10% MCDB	27.2	29.3	37.6	47.3	58.5	67.7	68.5	67.7	62.1	52.6	42.7	31.8	
	MDBR	13.6	13.6	12.9	13.2	14.3	14.3	12.7	12.7	12.8	11.0	10.8	12.0	
	5% DB	17.8	14.1	16.8	19.7	19.2	17.1	15.0	14.6	15.1	15.3	16.4	14.1	
(36) Mean Daily Temperature Range	MCDBR	18.1	12.9	13.3	13.1	11.2	10.1	9.2	9.2	9.5	10.3	15.0	13.4	
	MCWBR	18.0	14.9	16.3	18.2	16.9	14.8	13.3	12.9	12.9	14.8	16.3	13.9	
	MCWBR	18.0	14.1	14.6	14.4	12.2	10.8	9.3	9.4	9.6	11.4	15.9	13.5	
	MCWBR	18.0	14.1	14.6	14.4	12.2	10.8	9.3	9.4	9.6	11.4	15.9	13.5	
(40) Clear-Sky Solar Irradiance	taub	0.236	0.258	0.286	0.314	0.332	0.370	0.375	0.358	0.328	0.297	0.277	0.252	
	taud	2.424	2.329	2.399	2.458	2.460	2.326	2.316	2.370	2.460	2.564	2.537	2.425	
	Ebn at Noon	287	297	303	301	297	284	281	283	284	281	268	266	
	Edh at Noon	22	29	32	33	34	39	39	36	30	23	20	20	
(44) All-Sky Solar Radiation	RadAvg	452	742	1125	1374	1593	1755	1764	1568	1253	725	441	316	
	RadStd	55	82	87	160	163	137	137	80	102	68	57	28	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+108	N/A